

ICT-Based Research, Innovation, and Scientific Paper Writing for Professional Engineers

BY

PRADIP PAUDYAL, PHD

Group Activity

Research Problem Identification

Each member generates 5 research topics based on real-world problems or research gaps.

Follow the process:

Problem Identification → Gap Analysis → Research Question → Research Topic

Discuss and select one final topic as a group.

Justify the selection: “Why this topic and not the others?” considering relevance, feasibility, novelty, and impact.

List of topics for each group

G1:

G2:

G3:

Is the Topic Manageable?

Ask yourself:

Am I truly interested in this topic?

Does it meet all the assignment/personal requirements?

Do I have access to enough information?

Is a paper on this topic feasible?

Literature Review

A literature review is **not**:

- just a summary of sources
- a grouping of broad, unrelated sources
- a compilation of everything that has been written on a particular topic
- literature criticism (think English) or a book review

So, what is then?

- ✓ An integrated and critical analysis of previous research
- ✓ Directly related to the research question/topic
- ✓ Used to provide background and identify research gaps
- ✓ A foundation for developing research objectives and methodology

Purpose of Literature Review

Before Topic Selection

- Helps identify suitable research ideas
- Provides understanding of existing research and trends
- Assists in selecting relevant research materials

After Topic Selection

- Supports the significance and originality of the research
- Demonstrates that the study is not simple repetition
- Helps justify the contribution of new knowledge

Review of Related Literature

- Identifies what has already been done
- Highlights research gaps and future directions
- Reviews only literature relevant to the research topic/questions

Importance of Literature Review

- Provides background and context of the study
- Identifies major themes, concepts, and methods
- Avoids duplication of previous research
- Helps interpret and strengthen research findings

Steps for Conducting a Literature Review

- Identify the research topic or problem for investigation.
- Define keywords and search relevant literature from journals, conferences, and databases.
- Read and critically evaluate the quality, relevance, and findings of existing studies.
- Organize and synthesize the collected information by themes, methods, or trends.
- Identify research gaps, limitations, and opportunities for new research.
- Write and present the literature review with proper citations, highlighting the purpose and significance of the proposed research.

- As you write your review, consider these ways of expressing your ideas:
 - **Compare and contrast** views of different authors.
 - **Criticize** previous work.
 - **Highlight** gaps in existing research.
 - **Show** how your work relates to previous work.
 - **Identify** problems, conflicts, debates, gaps.
 - **Define** a research area in a new way.
 - **Question** previous results.

<https://researchguides.case.edu/c.php?g=157847&p=1036130>

Literature Resources

Books

Manuals

Journals

Conference and Workshop Proceedings

Reports

Newspapers, magazines, radio and television

Art exhibition catalogues

Multimedia 'literature'

Resource catalogues and online databases

Academic gateways (also called portals)

People

The Internet and Literature Reviews

Some highly rated journals in IS and computing

- *ACM Computing Surveys*
- *Communications of the ACM*
- *Digital Creativity*
- *IEEE Transactions*
- *Information Systems Research*
- *MIS Quarterly*



Google Scholar

Literature Resources and Techniques

Relevant literature may take several forms including **books, book chapters, published journal articles, conference papers or presentations, theses or doctoral dissertations.**

When choosing sources consider:

- Is it scholarly work?
- Does it have application to the topic?
- Is it based on theory?

Is the Source Trustworthy?

Researchers may want to consider the following questions when examining existing literature:

- ✓ Is the author an expert?
- ✓ Is the source current?
- ✓ Is the source complete?
- ✓ Is the source biased or unbiased?
- ✓ Is the source accurate and logical?

Conducting a literature review

Searching

Concept 1	Concept 2	Concept 3	Concept 4
Attitudes	Advertising	Mobile phones	Text messaging

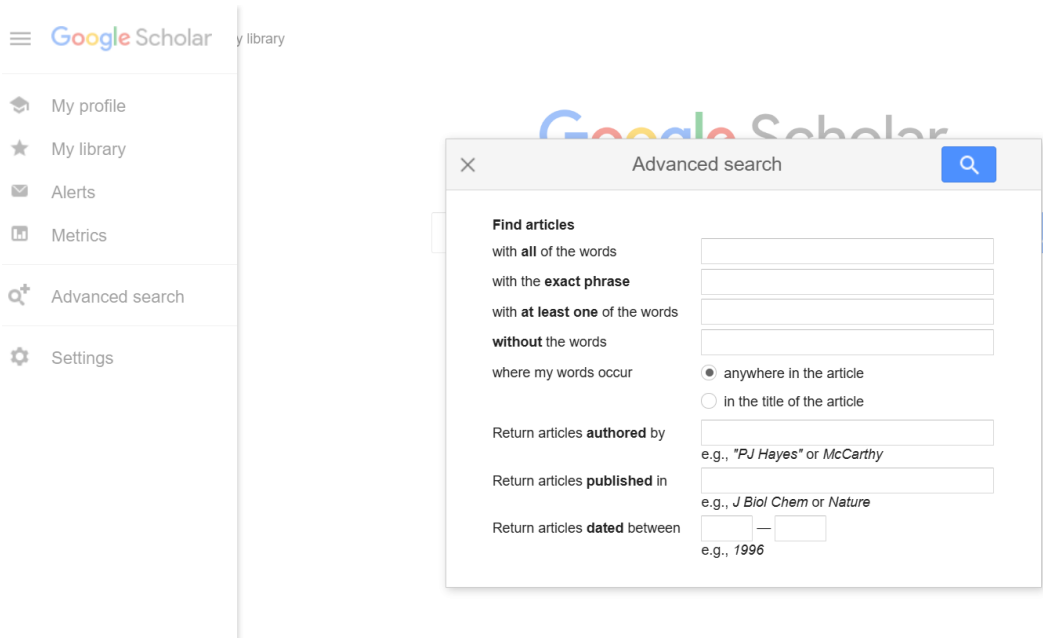
Figure 6.1 Splitting a research topic into separate concepts

Concept 1	Concept 2	Concept 3	Concept 4
Attitudes	Advertising	Mobile phones	Text messaging
Beliefs	Marketing	Cell phones	SMS
Prejudices	Selling	Telephones	

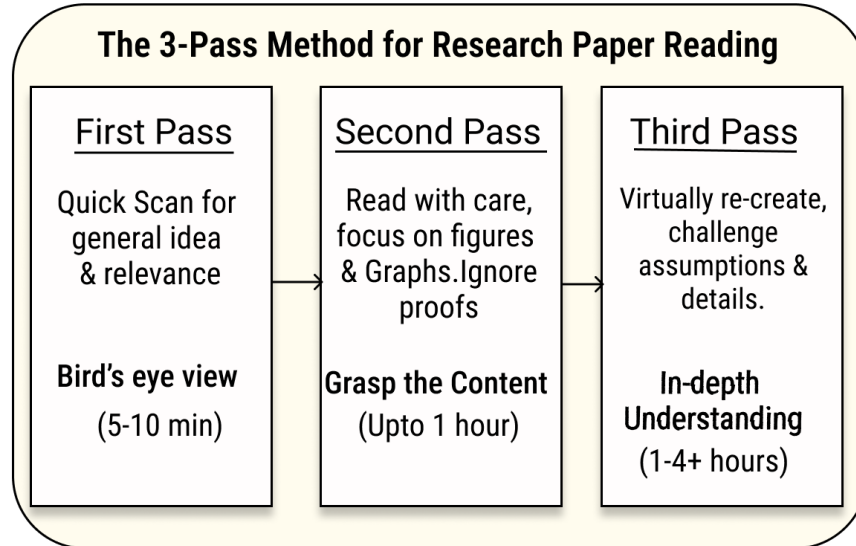
Figure 6.2 Alternative terms for research concepts

Example	Outcome
attitud*	Results contain words such as attitude, attitudes, attitudinal
mobile AND phones	Results contain both terms, but they do not necessarily occur together
mobile OR phones	Results contain at least one of these terms
'mobile phones'	Results contain the phrase 'mobile phones'

Figure 6.3 Using symbols and Boolean operators in a search



Conducting a literature review



How to Read Research Papers 10x Faster: The Three-Pass Method

@emmanuelseklevs

- 1st Pass: Bird's Eye View Test**
 - Read title, abstract, introduction
 - Scan section headings only
 - Read the conclusions
 - Glance at references

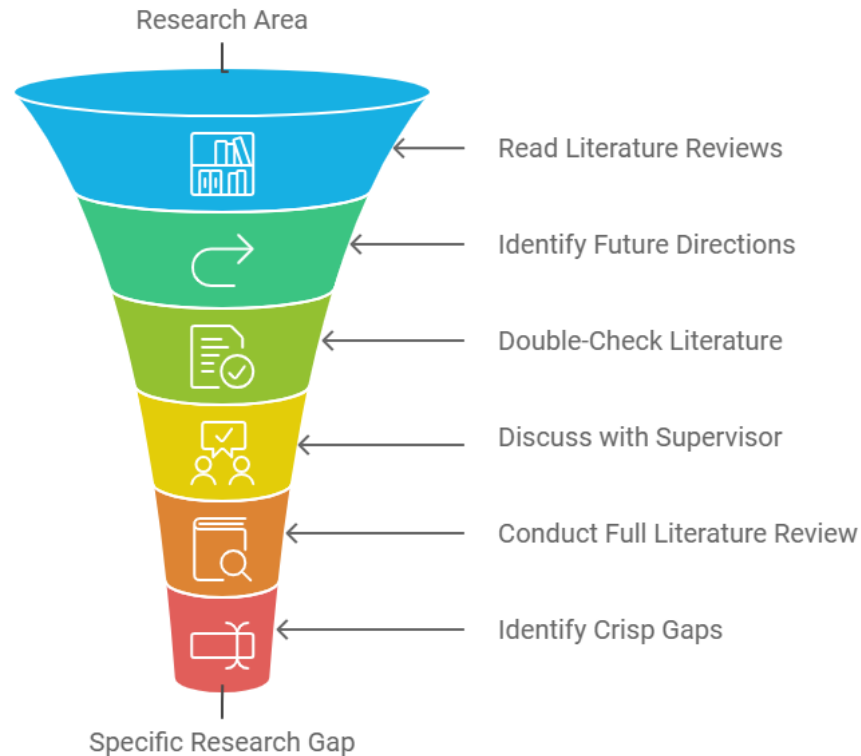
✓ Decide if paper is worth your time
- 2nd Pass: Content Grasp**
 - Study figures, diagrams, graphs
 - Mark key points in margins
 - Note unread references to follow up
 - Ignore proofs and details

✓ Summarise main thrust to someone else
- 3rd Pass: Deep Understanding**
 - Walk through the methodology step by step
 - Challenge every assumption
 - Identify hidden weaknesses
 - Note ideas for future work

✓ Reconstruct structure from memory

Identifying problem: Research question / problem statements and hypothesis

How to Identify Research Gap?



- ✓ When looking for **research gaps across publications** you may have noticed sentences like:
 - ...has/have not been... (studied/reported/elucidated)
 - ...is required/needed...
 - ...the key question is/remains...
 - ...it is important to address...
- ✓ These expressions often indicate gaps; issues or topics related to the main question that still hasn't been subject to a scientific study. Therefore, it is important to take notice of them: who knows if one of these sentences is hiding your way to fame.

How to find a Gap in Research

01

Select a topic or question that motivates you

02

Find keywords and related terms to your selected topic

03

Look for topics or issues that are missing or not addressed within (or related to) your main topic

04

Read systematic reviews

Elsevier Author Services Blog



Author Services

Conducting a literature review

Writing a critical review

gathering and presenting evidence to support your claim that you have created some new knowledge, specifically that:

- the topic is worthwhile;
- the research does not merely repeat the work of others (or if it does, there is a deliberate reason for this);
- You have created some new knowledge that was not known before.

Articles	Concepts			
	A	B	C	...
Adkins (1996)	*	*		
Bell (2000)			*	
Jones (1995)	*			
Smith (1996)		*		
...				

Figure 6.4 Using a matrix to map papers to concepts

Literature review: Making a reading grid

Source: <https://bit.ly/2E7MF68>

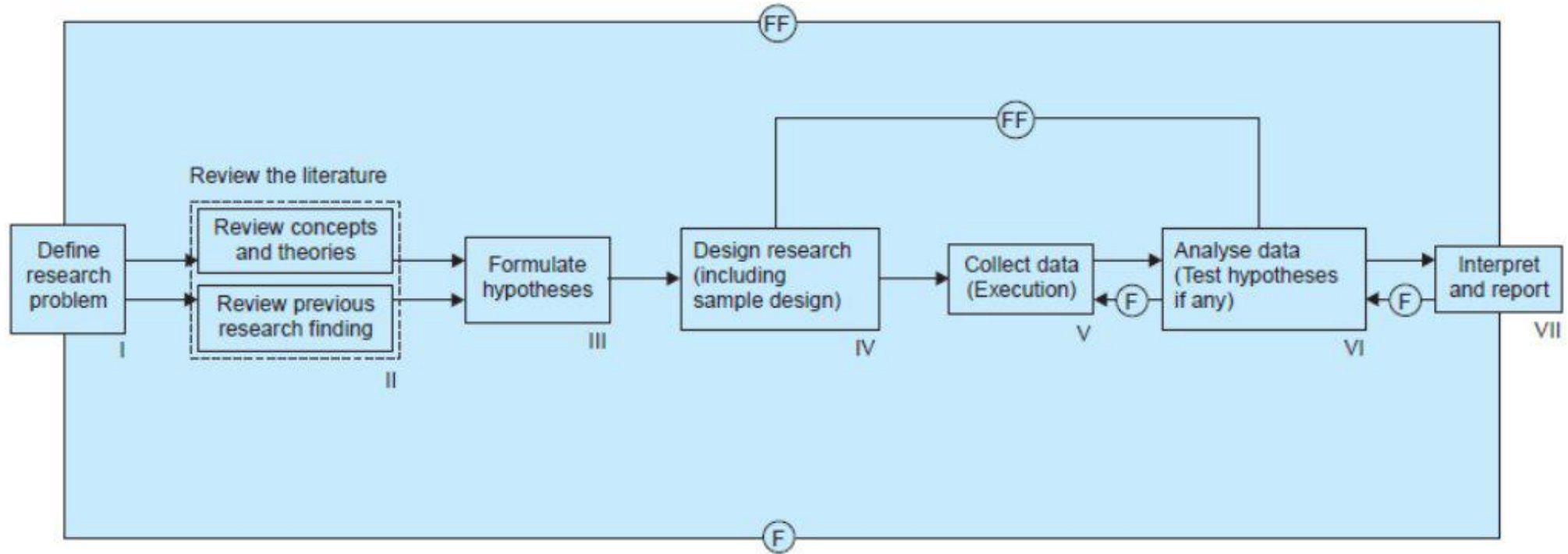
Source information	Junco (2015)	Karpinski, Kirschner, Ozer, Mellot & Ochwo (2013)	Kirschner & Karpinski (2010)	Lau (2017)
Research topic/question	Relationship between time on various FB activities and GPA	Multitasking impact on relationship between FB & academic achievement	Are there differences in academic performance of student FB users and non-users?	Non-academic FB use bad for GPA. Academic FB use good for GPA. Multitasking negative.
Methodology	• Survey: quant > USA > 1399 undergrad sts	• Survey: quant & qual > USA & Europe > 590 undergrad sts > 285 grad sts	• Survey: quant & qual > USA > 102 undergrad sts > 117 postgrad sts	• Survey: quant > Hong Kong > 348 undergrad sts
Findings	• FB bad for academic performance • Poss. good for engagement	• Negative relation between FB use and GPA • Negative effects of multitasking	• FB users lower GPA • Similar online time for FB & non FB users • Multitasking negative	• Non-acad. FB use & multitasking negative for academic performance
Limitations	• Cross-sectional & correlational • Focus on GPA	• Correlational • Focus on GPA	• Correlational • Acad./learning use of FB not considered	• Cross-sectional • Multitasking activities unclear
Areas for future research	• Engagement impact & FB use with high acad. performance?	• Focus on how FB is used by high performing sts?	• Focus on how FB is used by high performing sts?	• More focus on multitasking & high acad. performance?

Adapted from Auckland University of Technology (Library)

@OpenAcademics

- ✓ Create an outline and break the review into manageable sections.
- ✓ Organize literature logically using themes, theories, methods, or relevance.
- ✓ Critically analyze previous studies and relate them to the current research.
- ✓ Demonstrate the importance of the topic and identify research gaps.
- ✓ Use proper citations and references consistently.
- ✓ Present a clear research question that contributes new knowledge.

Research Process



Where (F) = feed back (Helps in controlling the sub-system to which it is transmitted)
(FF) = feed forward (Serves the vital function of providing criteria for evaluation)

Research problem

The key effort is to make the reviewers/readers understand

- What is the problem
- Why is the problem important

Problem statement answers one primary question: *Why* do this project?

- The emphasis is on the “why”

To answer this question, researcher must

- (1) define the problem and
- (2) document its existence

Research problem

The problem statement of the problem should quickly:

- summarize the problem from the vantage point of the readers,
- document its frequency and severity,
- show your familiarity with prior research or work on the problem, and
- justify why this problem should be investigated

Do not assume that everyone sees the problem as clearly as you do

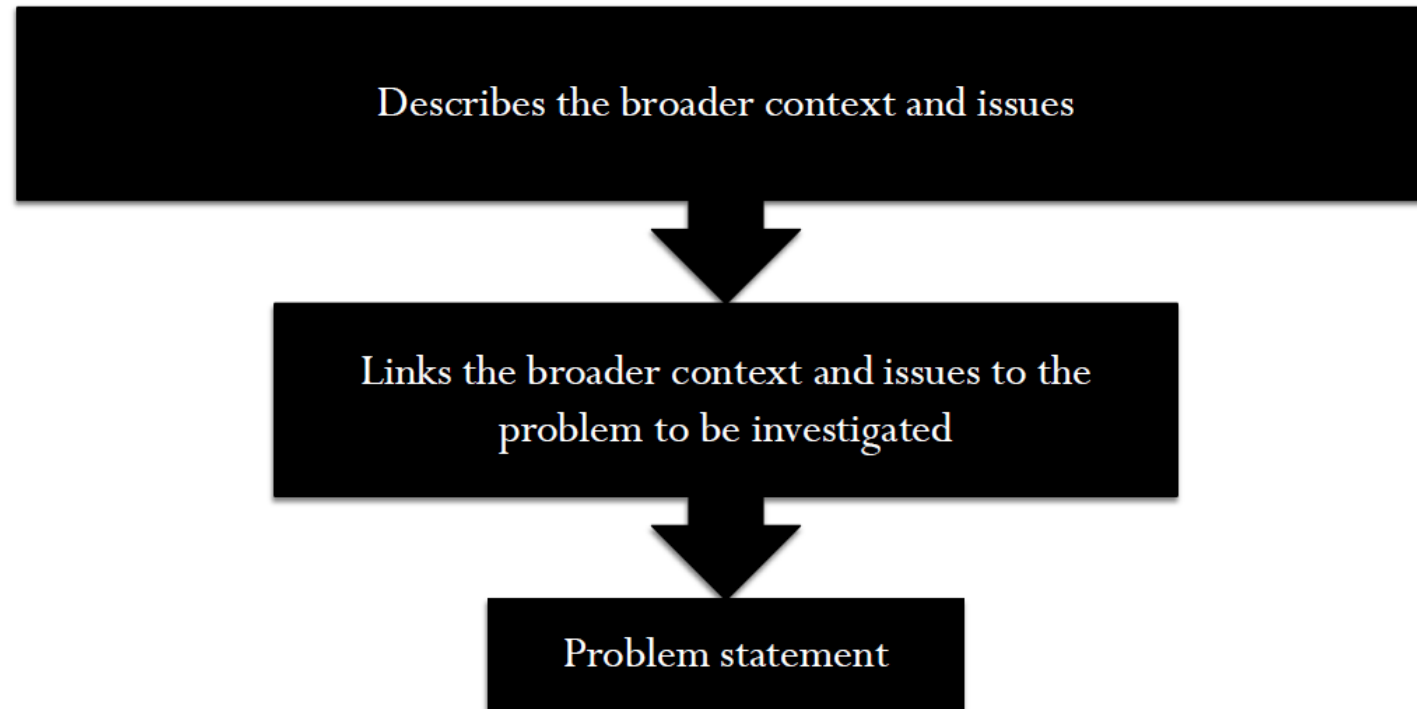
- even if the problem is obvious, your readers want to know how clearly you can state it

1.2 Problem Statement

The focus of this thesis is on introducing network services rather than using or adapting existing network services. Introducing network services is no new activity, this can already be done with a GPL. But since every technically possible problem can be solved with a GPL, there is no clear notion of the mechanisms that are required to introduce network services. It is expected that by developing a DSL, it will become clear which mechanisms are essential or common to introduce network services.

Let's Evaluate

Research problem



Research problem

Research is extending the knowledge frontier

- That is, building upon other researchers' work

Cite other people's work (not only yours)

Explain

- The current limits and gaps in knowledge
- Your study's contribution to fill in current gaps in knowledge or to extend the knowledge frontier
- The broader impact or benefit to the society, country, or the world
- A good statement to have in your research: *"If successful, the benefits of this research would be ..."*

Objectives

Objectives describe the purpose of the study

Must be

- specific
- clear
- logical
- immediate (have a time frame)
- concise, and
- achievable (measurable)

Anticipate	Construct	Discriminate	Measure
Arrange	Contrast	Display	Motivate
Assemble	Coordinate	Distinguish	Organize
Assess	Decrease	Establish	Quantify
Build	Demonstrate	Estimate	Solve
Categorize	Describe	Evaluate	Stimulate
Classify	Design	Explain	Summarize
Compare	Detect	Illustrate	Translate
Conduct	Discover	Increase	

May have two sections: general and specific objectives

Or split study into parts/sections, and for each section a statement of its study objectives

- But ensure the sections are related to one another to solve or investigate the overall problem

Hypothesis to be tested

- Expected results
- What you expect/believe the results would be

Use action verbs in objectives such as:

Group Activity

Groups search papers, extract gaps, summarize findings, prepare references

Feedback, topic refinement, and feasibility discussion

Deliverables:

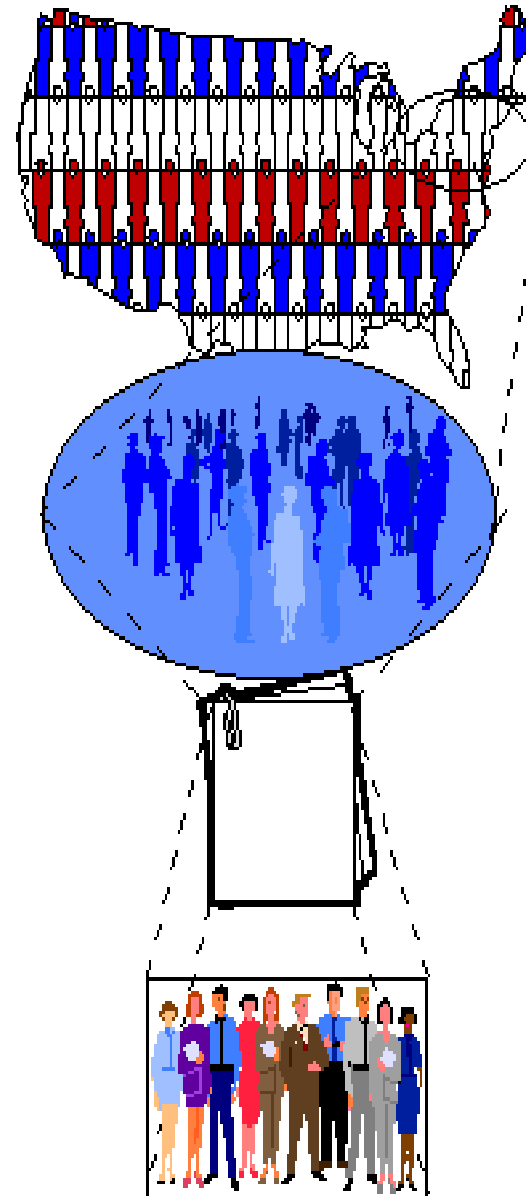
Research title/topic, problem statement, initial literature review summary and identified research gap

Who do you want to generalize to?

What population can you get access to?

How can you get access to them?

Who is in your study?



The Theoretical Population

The Study Population

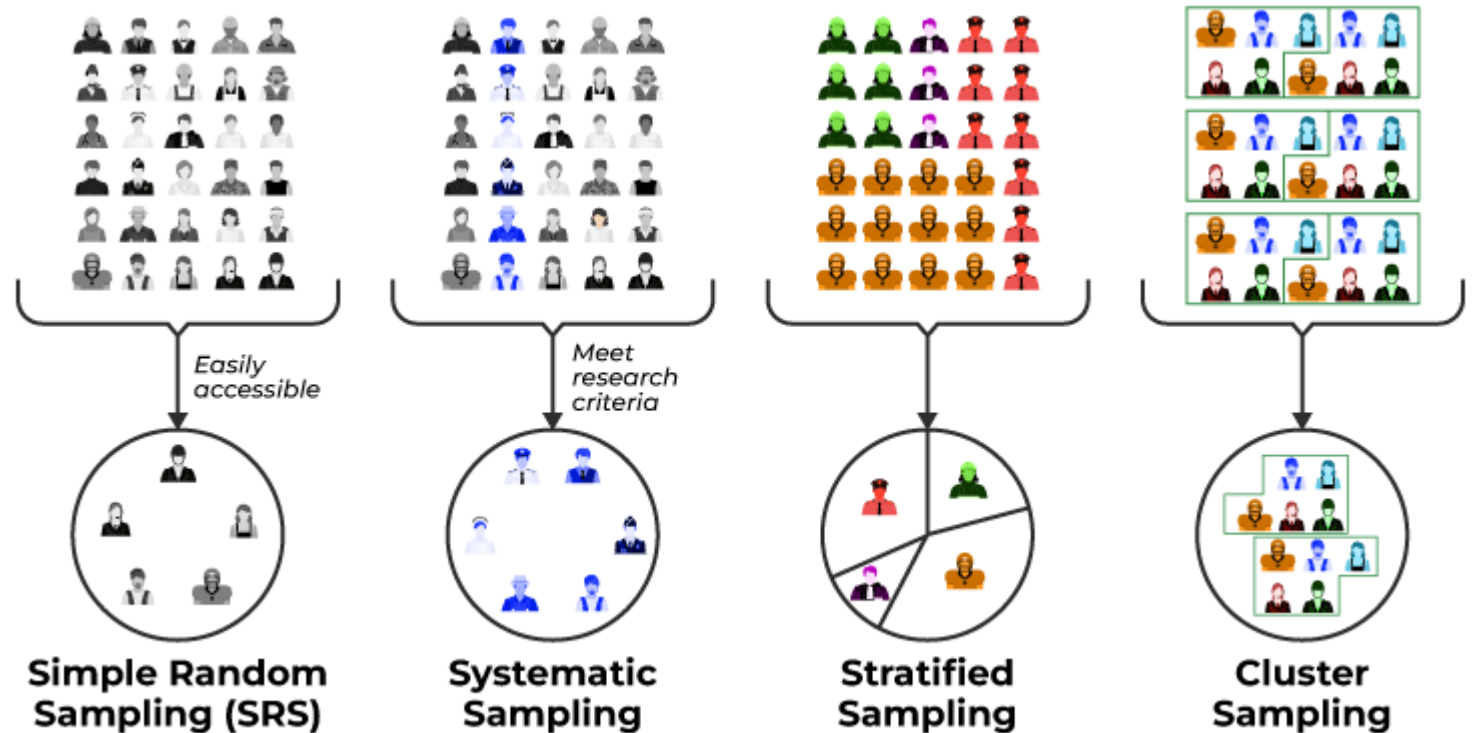
The Sampling Frame

The Sample

PROBABILITY SAMPLING.....

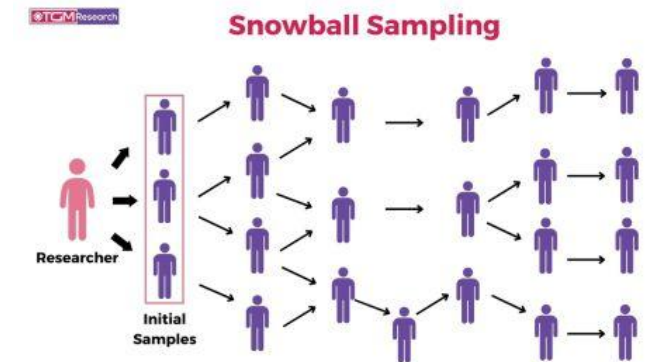
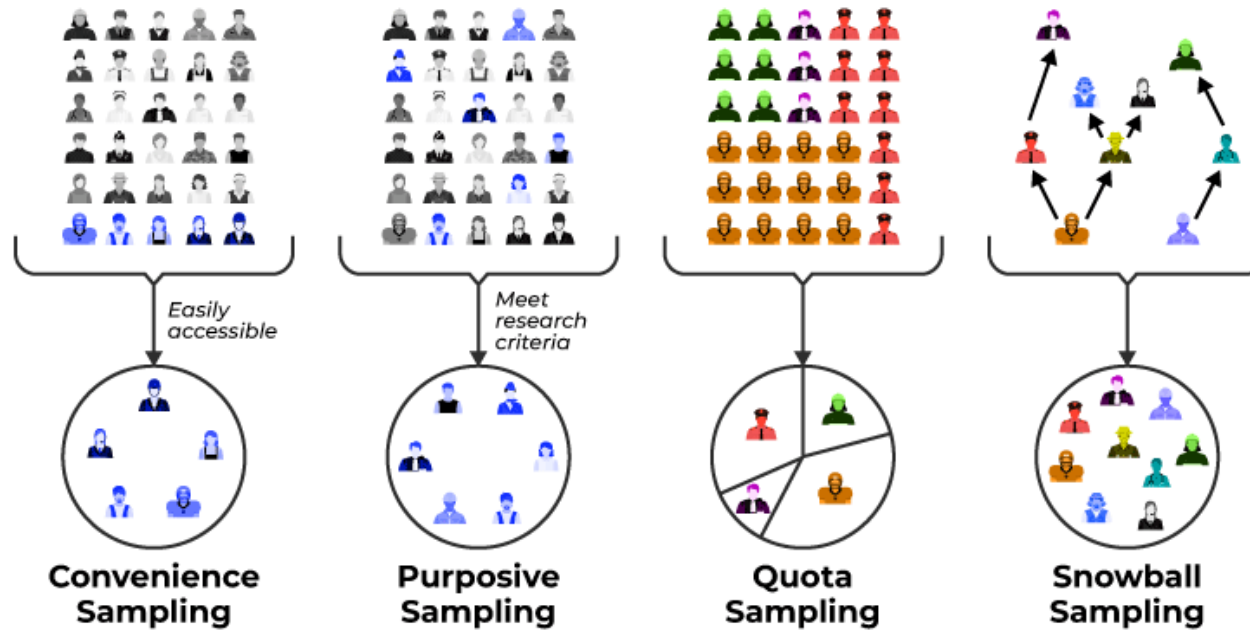
Probability sampling includes:

- Simple Random Sampling,
- Systematic Sampling,
- Stratified Random Sampling,
- Cluster Sampling



NONPROBABILITY SAMPLING.....

- Nonprobability Sampling includes: [Accidental Sampling](#), [Quota Sampling](#) and [Purposive Sampling](#). In addition, nonresponse effects may turn any probability design into a nonprobability design if the characteristics of nonresponse are not well understood, since nonresponse effectively modifies each element's probability of being sampled.



Research Design

A research design is a **plan** according to which observations are made and data assembled. It provides the empirical and logical basis for drawing conclusions and gaining knowledge.

Why?

- Provides the blueprint for research
- limits the boundaries of research activity
- enables investigation to anticipate potential problems

(Balck and Champion, 1976)

Ahuja (2007) explains that the research designing proceeds in eight phases as under:

- 1) **Stating the problem**, i.e., to show the type of research it will be, and to what extent will it contribute to the academics and understanding of society.
- 2) **Review of other studies**, i.e., going through the findings, hypotheses or theories developed by other scholars in one's own discipline or other disciplines.
- 3) **Operationalising concepts**, i.e., giving specific meanings to the terms used.
- 4) **Identifying variables of the study**, i.e., pointing out key variables (see BOX) in the study and methods of measurement.
- 5) **Fixing sample**, i.e., deciding the numbers of subjects from whom the data is to be collected and how these subjects are to be chosen.
- 6) **Specifying tools of study**, i.e., whether data are to be obtained through questionnaires, schedule, interview or observation. Whether it will be case study, survey study, field study or experimental study.
- 7) **Designing the kind of analysis**, i.e., whether quantitative or qualitative analysis for the study.
- 8) **Fixing time-schedule and budget**, i.e., how much time and budget is to be devoted to the study.

Major Research Design Types in ICT

Research Design	Purpose / Description	Example Research Objectives
Experimental Research	To test hypotheses and evaluate the performance of algorithms, systems, protocols, or hardware under controlled conditions. This is one of the most common designs in Computer Engineering and Computer Science research.	Compare the performance of Transformer and CNN models for image classification. Evaluate the efficiency of post-quantum cryptographic algorithms. Assess the performance of RISC-V and ARM processors for AI workloads. Compare routing protocols for 6G-enabled IoT networks.
Design Science Research (DSR)	<i>To create and evaluate innovative artifacts such as algorithms, frameworks, software systems, architectures, hardware prototypes, or cybersecurity solutions. Widely used when proposing a novel solution.</i>	<i>Design and develop an AI-powered educational assistant using Large Language Models. Develop a blockchain-based healthcare data-sharing framework. Design an intelligent intrusion detection system for cloud environments. Develop an Edge AI architecture for autonomous vehicles.</i>
Simulation-Based Research	To study complex systems through modeling and simulation when real-world experimentation is expensive, time-consuming, or impractical. Common in networking, communication, embedded systems, and smart infrastructure research.	Simulate UAV swarm communication networks to evaluate routing efficiency. Model smart grid resilience under cyberattack scenarios. Evaluate resource allocation strategies in edge-cloud computing environments. Simulate vehicular communication networks for intelligent transportation systems.

Major Research Design Types in ICT

Empirical/Data-Driven Research	To discover patterns, relationships, and evidence through analysis of real-world datasets, software repositories, logs, sensor data, or user-generated data. Increasingly important in AI and software engineering research.	Analyze software development practices using GitHub repositories. Investigate factors affecting software defect occurrence. Analyze cybersecurity incident datasets to identify attack patterns. Study user behavior in online learning platforms using educational data mining.
Survey Research	To collect perceptions, experiences, attitudes, or practices from users, professionals, organizations, or stakeholders through questionnaires and structured instruments. Common in HCI, software engineering, cybersecurity awareness, and technology adoption studies.	Assess AI adoption among software developers. Investigate cybersecurity awareness among university students. Examine factors influencing cloud computing adoption in enterprises. Evaluate user acceptance of smart healthcare applications.
Case Study Research	To conduct an in-depth investigation of a technology, organization, deployment, or project within a real-world context. Useful for understanding implementation challenges and lessons learned.	Investigate AI implementation in healthcare institutions. Examine smart campus deployment in higher education. Analyze enterprise cloud migration strategies. Evaluate Industry 5.0 adoption in manufacturing systems.

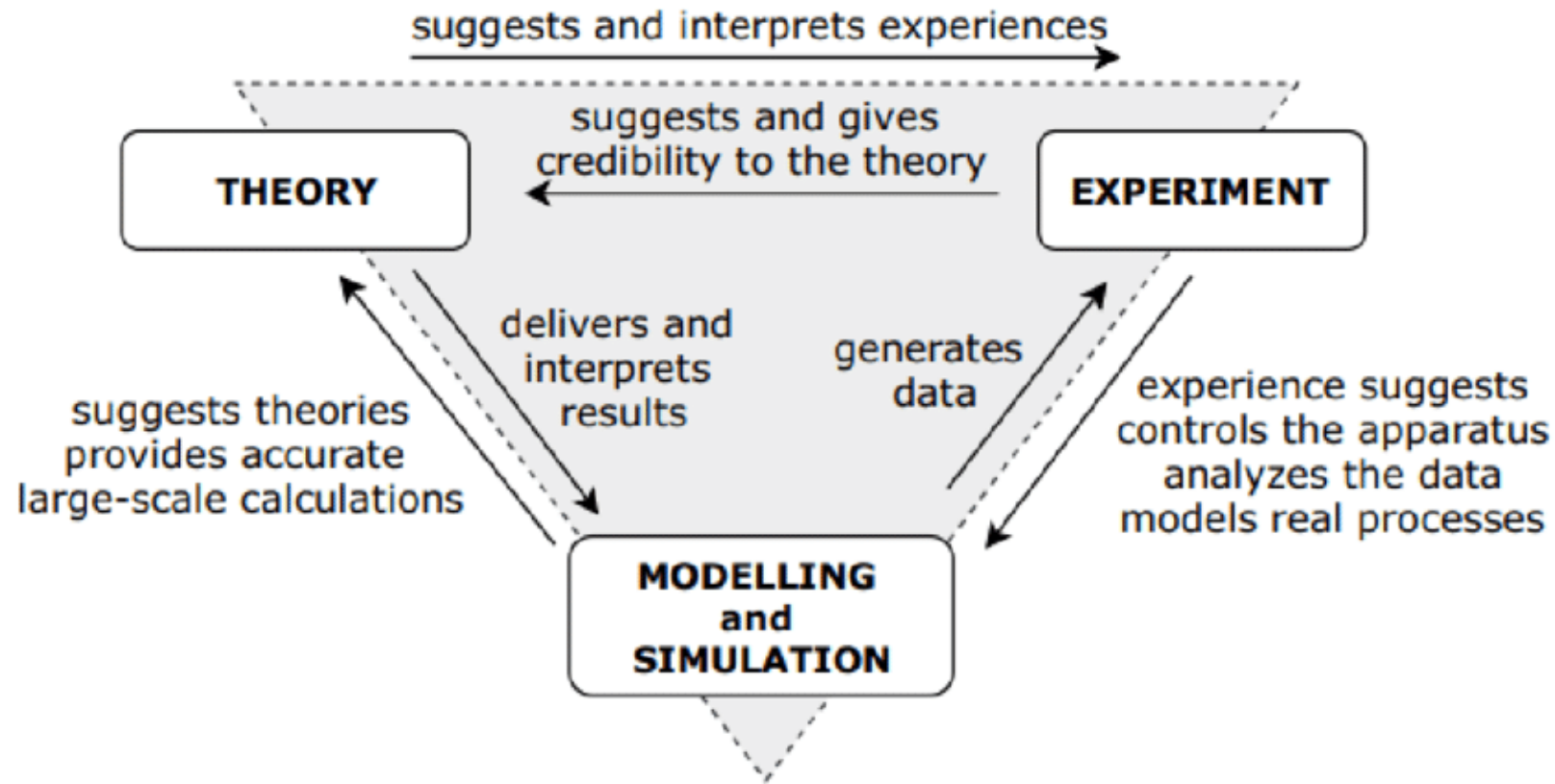
Discussion Points

Research Topic: AI-Based Intrusion Detection System

Possible Research Objectives under Different Research Designs:

- **Exploratory:** Explore the challenges of applying Generative AI to intrusion detection.
- **Descriptive:** Analyze attack patterns in enterprise networks.
- **Explanatory:** Determine the effect of feature selection on detection accuracy.
- **Predictive:** Predict cyberattacks using deep learning models.
- **Design Science:** Design and develop a novel intrusion detection framework.
- **Experimental:** Compare multiple AI algorithms for intrusion detection performance.
- **Simulation:** Simulate cyberattacks in a virtual network environment.
- **Case Study:** Evaluate the deployment of an intrusion detection system in a real organization.

Simulation-based Research



Design Science Research

- **Design Science Research (DSR)**

= Research paradigm in information systems/ICT research (also used in other disciplines, such as SWE)

- „extend the boundaries of human and organizational capabilities by creating new and innovative artifacts“

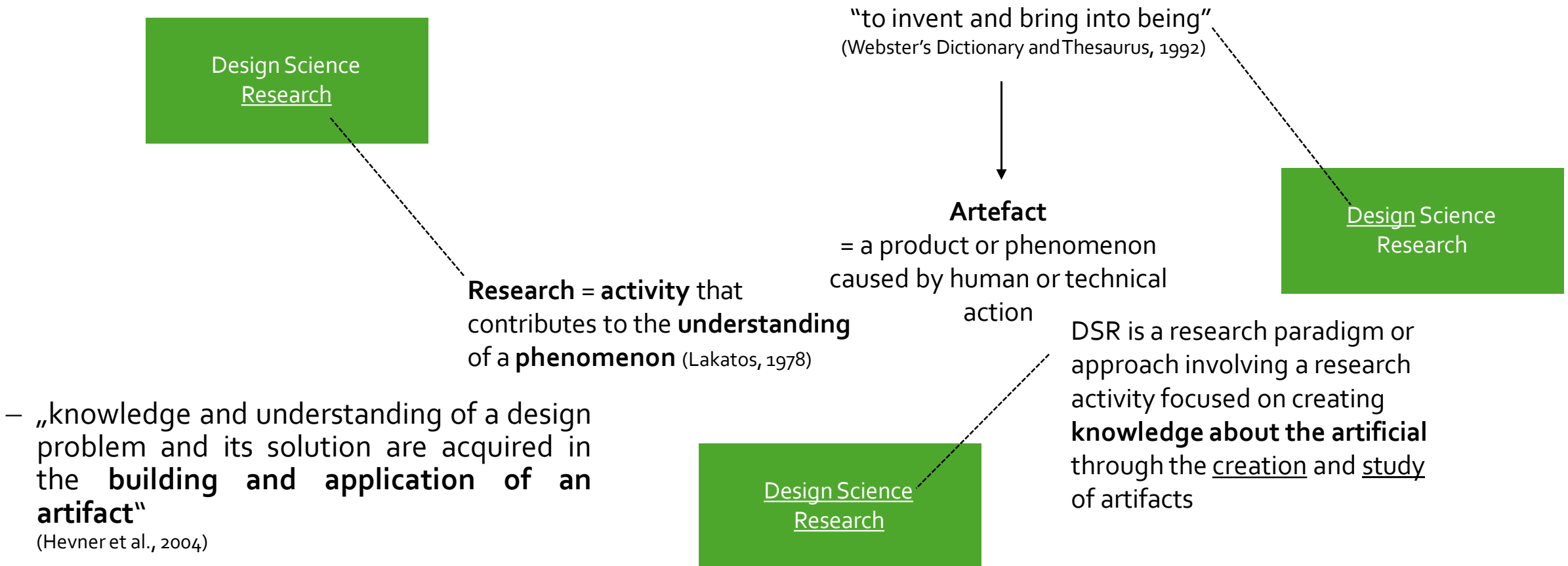
(Hevner et al., 2004, p. 75)

- → "exploring by building"

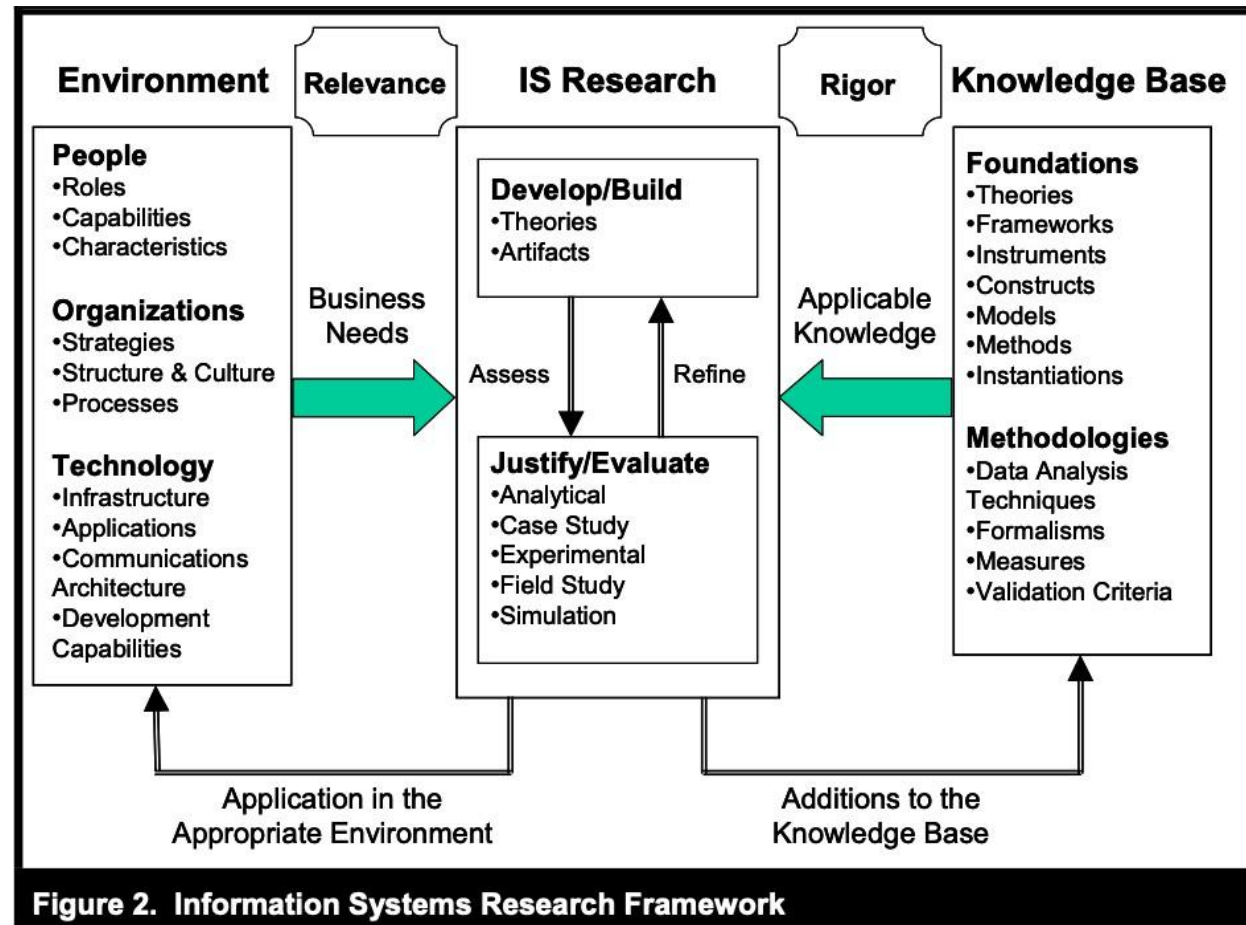
(Vaishnavi and Kuechler, 2015, p. 2)



Design Science Research



Information Systems Research Framework



Three Cycle View

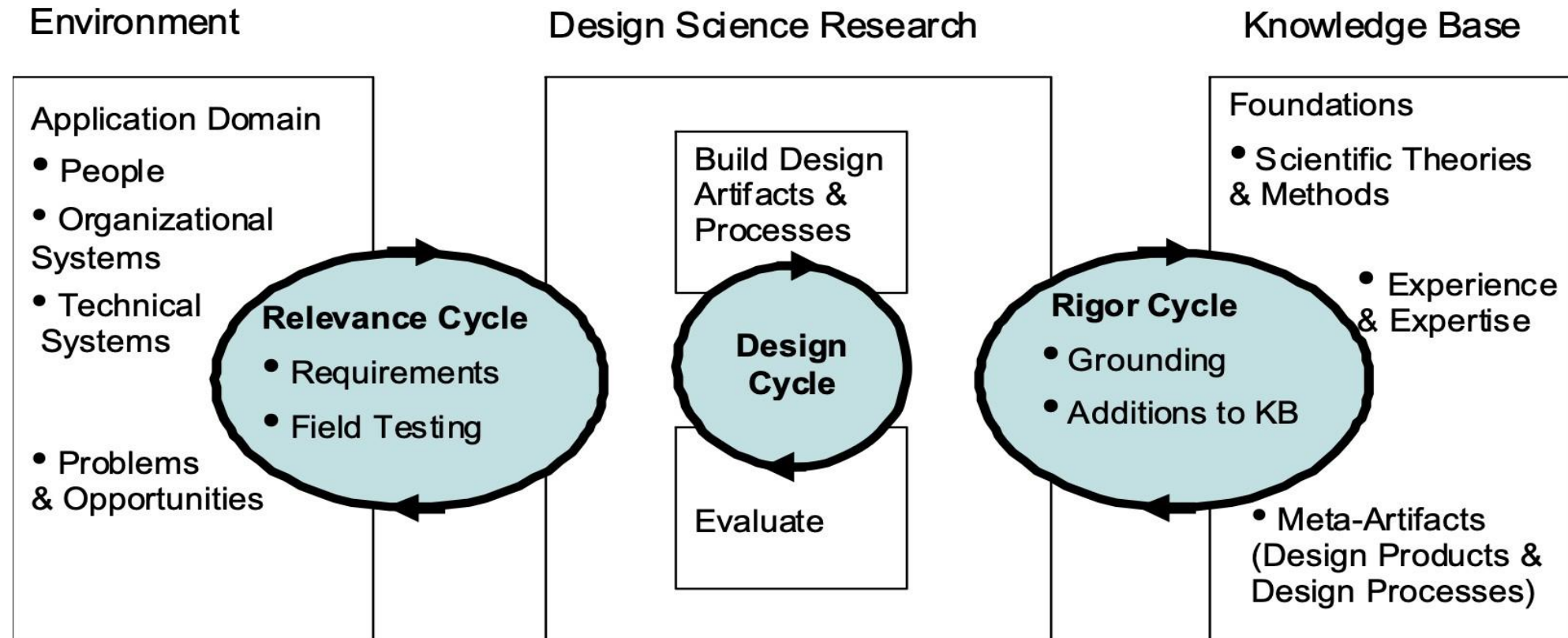
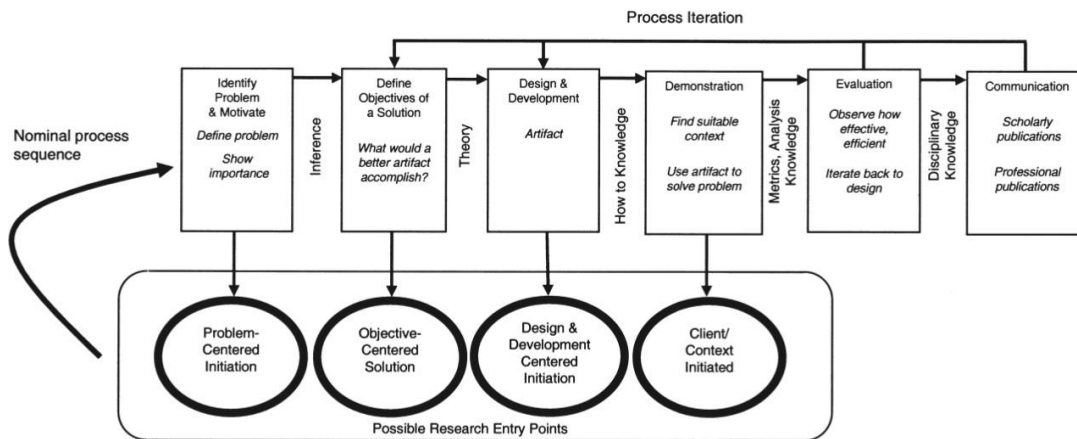


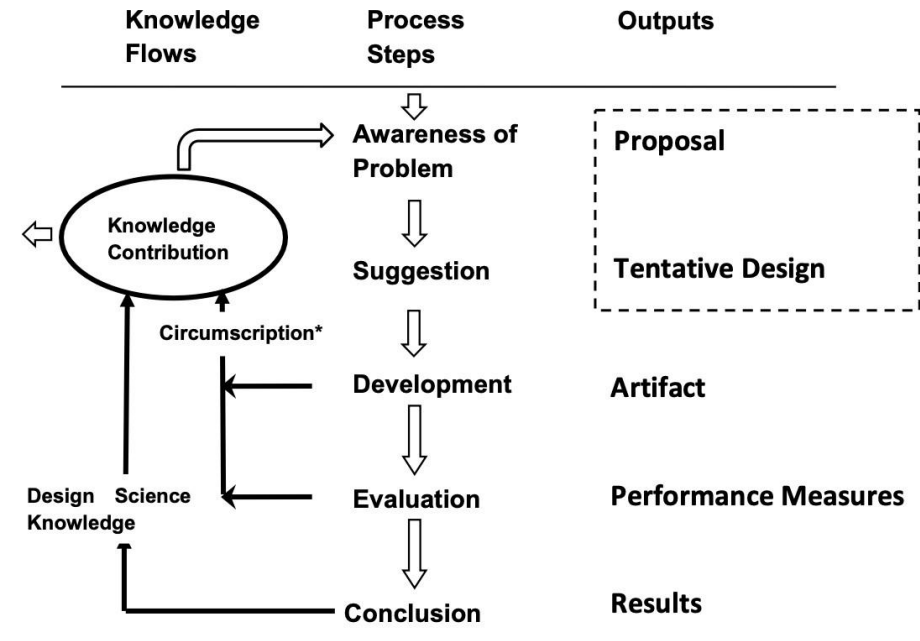
Figure 1. Design Science Research Cycles

DSR Frameworks

Design Cycle



Design Science Research Methodology (DSRM)
(Peppers et al., 2007)



General Methodology of Design Science Research (GMDSR)
(Vaishnavi and Kuechler, 2007)

Design Science Research (DSR)

Design Science Research (DSR) is a problem-solving paradigm that creates and evaluates innovative artifacts (e.g., algorithms, software, methods, or models) to solve real-world problems. It bridges the gap between theory and practice by producing both a functional solution and generalizable design knowledge.

The methodology is typically structured into an iterative, six-step process:

1. Problem Identification and Motivation

Define a specific, real-world organizational or technical problem. Quantify the problem's value and explain why a new artifact is required to solve it.

2. Objectives of a Solution

Translate the problem into measurable objectives. These goals determine whether the proposed solution is better than existing alternatives and serve as the criteria for later evaluation.

DSR

3. Design and Development

Create the artifact. Depending on your field, this could take the form of:

- **Constructs:** Vocabulary or terminology (e.g., entity-relationship models).
- **Models:** Sets of propositions or statements expressing relationships (e.g., financial data streams).
- **Methods:** Algorithms or guidelines used to perform tasks (e.g., software debugging process).
- **Instantiations:** Implemented or working systems (e.g., a software prototype).

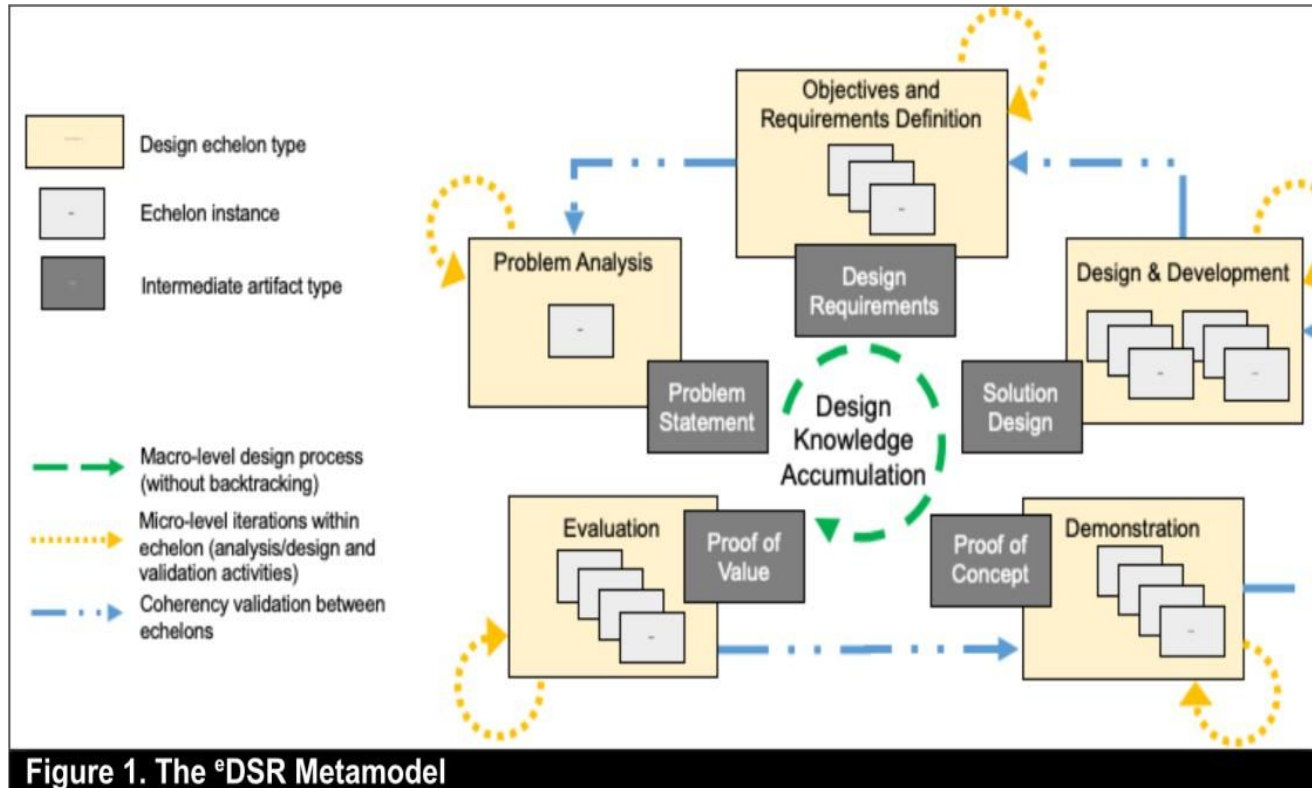
4. Demonstration

Apply the artifact to one or more real-world instances to show that it works. This can be achieved through case studies, simulations, or controlled experiments.

5. Evaluation

Observe and measure how well the artifact supports the solution. Compare the results against the objectives defined in Step 2. If the artifact fails to meet expectations, the design process must be iterated upon.

DSR Frameworks



Echeloned DSR (eDSR) methodology
(Tuunanen et al., 2024)

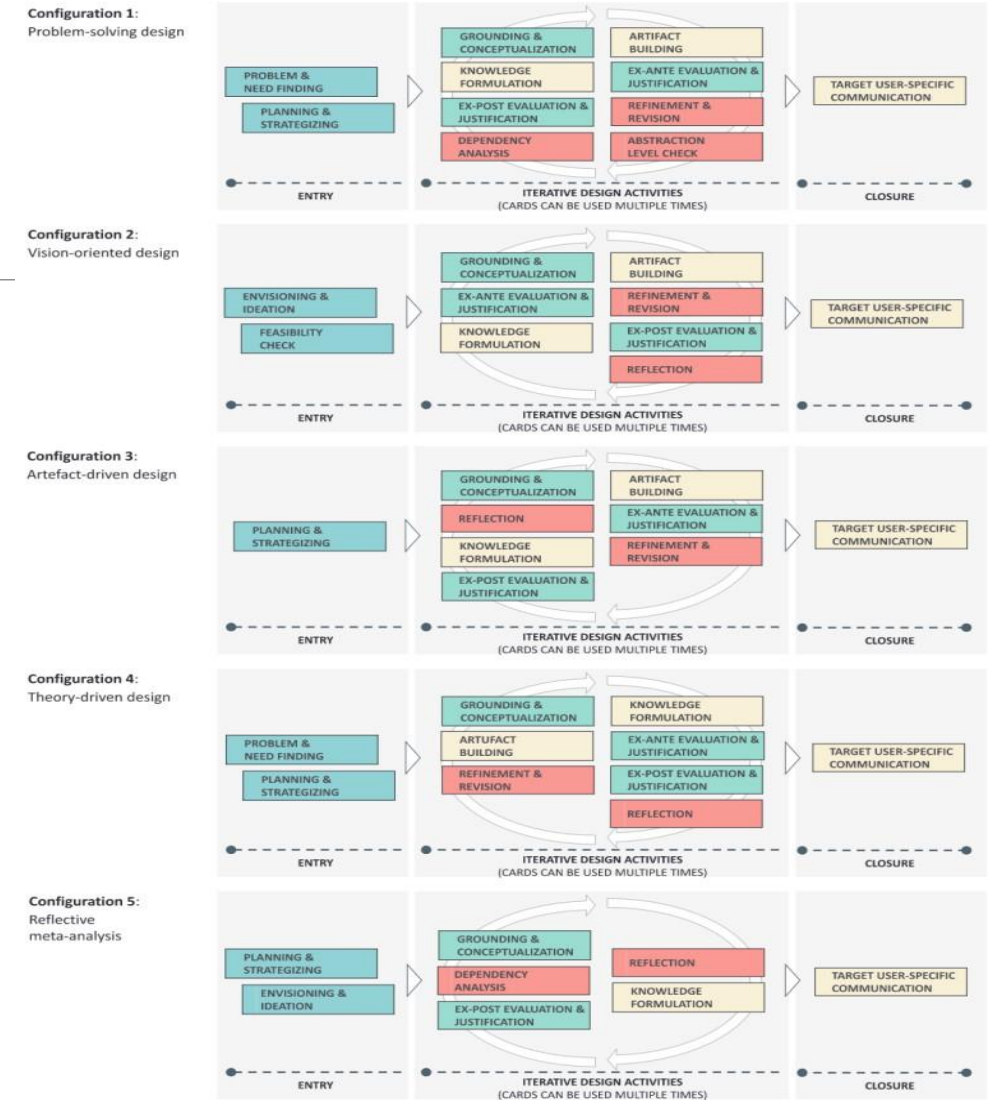


FIGURE 5 Overview of card configurations.

BAUSTEIN
(Schoormann et al., 2024)

What is a research proposal?

- A research proposal is a ...
 - A document that contains details about a scientific investigation to be carried out (*i.e.*, not started yet)
 - It contains details about:
 - The problem to be studied
 - How the investigation will be conducted
 - Expected results and contribution
 - Work schedule / Time frame
 - Budget (for those seeking funds)

Why do we need a research proposal?

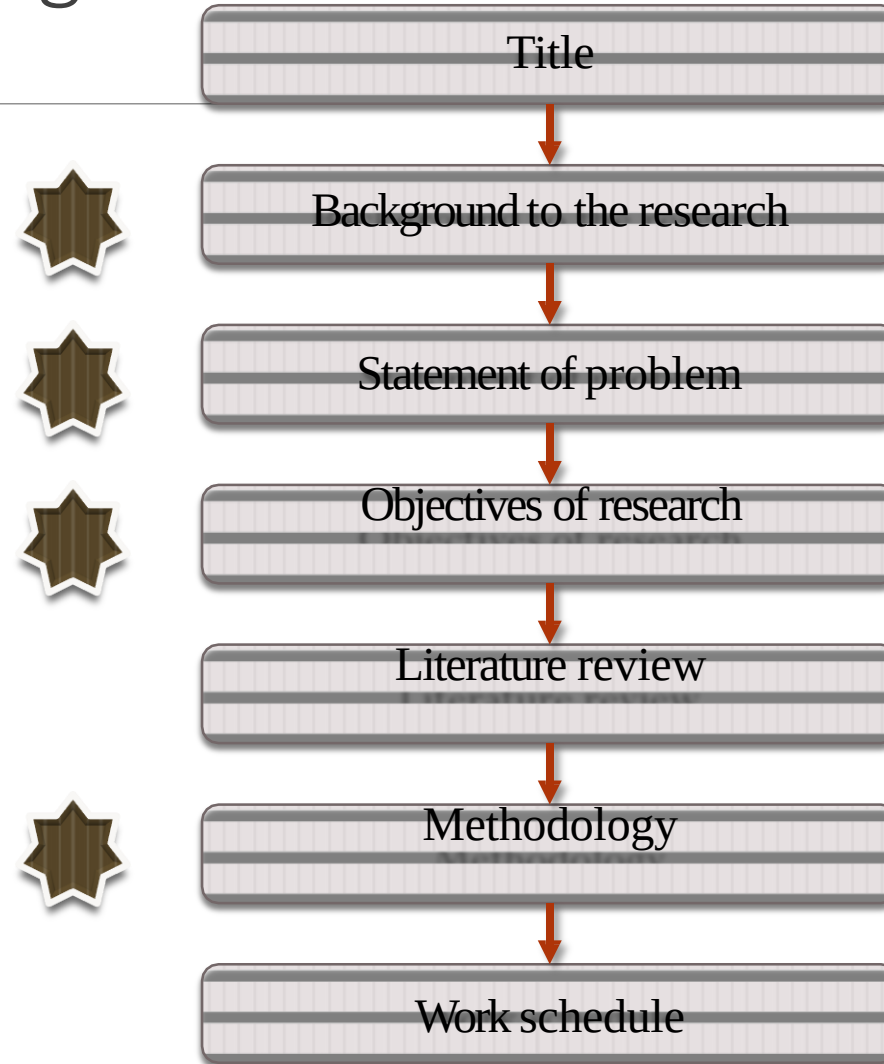
- A blueprint (plan) of the research program
- It forces us to contemplate in detail about
 - What we want to research
 - Why we want to do the research
 - How to do the research
 - What do we need to do the research
 - Are we able to do the research

Main purpose of a research proposal

- It is to show that
 - the problem you propose to investigate is significant enough to warrant the investigation,
 - the method you plan to use is suitable and feasible, and
 - the results are likely to prove fruitful and will make an original [or significant] contribution.
- In short, what you are answering is 'will it work?,,

- *Uni. of Queensland, Australia*

Key ingredients of a research proposal



Title

- Good titles identify the field(s) of research and indicate the kind of results to be obtained
- Avoid
 - Too long titles
 - Too general or vague titles

Background to the research and problem statement

- The key effort is to make the reviewers/readers understand
 - What is the problem
 - Why is the problem important
- Problem statement answers one primary question: "Why do this project?"
 - The emphasis is on the "why"
- To answer this question, proposal writers must
 - (1) define the problem and
 - (2) document its existence

- Research is extending the knowledge frontier
 - That is, building upon other researchers' work
 - Cite other people's work (not only yours)
- Explain
 - The current limits and gaps in knowledge
 - Your study's contribution to fill in current gaps in knowledge or to extend the knowledge frontier
 - The broader impact or benefit to the society, country, or the world
 - A good statement to have in your proposal: *'If successful, the benefits of this research would be ...'*

Key questions to be answered in a proposal

- Does your proposal:
 1. Specify the conditions you wish to change?
 2. Define the gaps in existing programs, services, or knowledge?
 3. Include appropriate statistical data about the frequency and severity of the problem?
 4. Clarify what will happen if nothing is done about this problem?
 5. State the problem in terms of human needs, not your opportunities?
 6. Cite pertinent theoretical literature, research findings, or ongoing studies?
 7. Convey the focus of your project early in the narrative?
 8. Point out the relationship of your project to a larger set of problems or issues?

Research objectives

- Objectives describe the purpose of the study
- Must be
 - specific
 - clear
 - logical
 - immediate (have a time frame)
 - concise, and
 - achievable (measurable)

- Use action verbs in objectives such as

Anticipate	Construct	Discriminate	Measure
Arrange	Contrast	Display	Motivate
Assemble	Coordinate	Distinguish	Organize
Assess	Decrease	Establish	Quantify
Build	Demonstrate	Estimate	Solve
Categorize	Describe	Evaluate	Stimulate
Classify	Design	Explain	Summarize
Compare	Detect	Illustrate	Translate
Conduct	Discover	Increase	

- May have two sections: general and specific objectives
- Or split study into parts/sections, and for each section a statement of its study objectives
 - But ensure the sections are related to one another to solve or investigate the overall problem
- Hypothesis to be tested
 - Expected results
 - What you expect/believe the results would be

Literature review

- Important to
 - Identify,
 - Evaluate,
 - Interpret,
 - Assimilate,
 - and Summarize all studies relevant to your study
- Impartial review of all studies
- Do not put all past studies into your proposal!
 - Select those really important (and current) studies

Methodology

- Lab analysis methodology
 - Cite reference; do not elaborate on commonly used methods
- Site location and its general characteristics
- Experimental design and layout
 - Ensure robustness of statistical analyses
- Statistical methods for data analysis
- Equipment and software needed
- Data collection
 - When will it begin and end
 - Frequency of data collection
 - What and how will be the data be collected
 - Type of data to be collected
 - How much data are needed

Work schedule

- Describes the time table for your work
 - When you will begin and end each part of your study
- Create a table or Gantt chart to depict your schedule

	YEAR 1												YEAR 2											
ACTIVITIES	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC
Activity 1	x	x	x	x																				
Activity 2					x	x	x	x	x	x	x													
Activity 3							x	x	x															
Activity 4												x	x	x	x	x	x	x	x	x				
Activity 5																	x	x	x	x	x	x	x	x

References

- Choose and follow a specific reference format
- Use current/latest references (after year 2010)
 - Avoid old references unless they are key references or key studies

Cleaning up your research proposal

- Check your proposal's spelling and grammar
- Brevity is important
 - Your proposal is not a thesis or a journal research paper
 - Highlight important issues, no need to put all information into the proposal
- Number of pages?
 - Check with what is required (including format of proposal)
 - Usually only 5 to 6 pages
- Make sure your supervisory committee gets a copy of your proposal
 - Get their feedback, especially from your main supervisor